A Universe That Expands Because It Exists

A Plain Language Explainer of This Cosmology

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In brief. This is the story the papers tell, told without their formalism. It starts from a result in standard general relativity: a collapsing star's event horizon is not a surface that forms at some moment, but a single occurrence at the end of the outside world's time. Read with the roles of light and matter exchanged, the same geometry carries the collapse on as a new universe, and the one constant it contains, the cosmological constant, fixes that universe's whole expansion curve — the same curve astronomers measure. So the universe expands because of the shape it has, not because something pushed it. Its space is a closed sphere, and yet the slices its observers measure on are exactly flat. Between the parent's collapse and our expansion runs a passage, the lap, part of which is crossed in imaginary time. That crossing hands on the universe's size and erases its lopsidedness, which is why the sky is smooth, and it starts every sound wave together. The turn from a slowing to an accelerating expansion comes at 9.8 billion light-years, the radius at which the universe's own mass can no longer hold it — the same rule that decides whether a group of galaxies keeps its outermost members. The same geometry fixes matter's three families and their handedness, though not their masses. The explainer ends where the measurements stand: what fits, what is left over, and what the sky will judge next.

Why does the universe expand? Because it exists; and existing as it does, it couldn't do otherwise. That is the answer this collection of papers delivers, and what follows is the plain-language story of how it gets there. But an answer like that only means something against the one it replaces. In the standard picture the universe begins with an infinite rate of expansion and an infinite braking against it, and it survives only because the first infinity was large enough to outlast the second. The cosmological constant, the one ingredient that might have explained anything, contributes nothing at all until billions of years later. Asked why the universe expands, that picture is really answering a harder question badly: why it should exist at all.

In the cosmology I'll describe here the curve is the same — the same expansion law our telescopes measure — but nothing is supplied to keep it going. There is no initial push and no braking against it; there is one constant, present from the first instant, and the shape of a single fixed geometry read on a cosmic clock. Take that as the starting point, and the interesting question stops being why the universe expands. It becomes what the rest of it looks like: where it came from, what crossed its beginning, why it started speeding up when it did, and why the same radius that marks that moment also decides whether a group of galaxies holds together.

The horizon that never forms

Start with something familiar: a star collapsing into a black hole. The usual story has the star fall through its own event horizon, after which the horizon sits there as a permanent surface, a one-way door hanging in space.

But ask *when* the horizon forms, by the clock of anyone outside, and the answer is: never. Not at any moment of the outside universe's history, however late. A clock falling with the star reaches the horizon in a finite time of its own; the outside universe keeps every moment of its own history on its side of that event, and the event of the horizon itself lies beyond all of them, at the outside universe's end.

This is not a trick of light, of signals arriving late. It's a fact about how the geometry orders events. Every way of slicing the outside world into "nows" piles up against the horizon without ever reaching it. And it isn't a matter of which clock you pick, either: even the descriptions built to carry a traveller smoothly across assign the horizon's events no separation at all — distinct events, in a definite order, at one place and one instant. The horizon isn't a place that exists for some amount of time. It is a single occurrence, and it happens at the end of the outside world's time — the collapsing star drawing marginally closer to it at every moment leading up to that final occurrence, in any valid description.

None of this depends on the star being round, and the order those events come in is not arbitrary. A black hole that has settled down looks the same from one moment to the next, and that sameness in time is a symmetry of its geometry — for a spinning hole, sameness in time together with a steady

turning at the horizon's own rate. Each light path running in its horizon is one of that symmetry's own paths, so the events along it come in the order the symmetry counts them off. Any horizon built that way — physicists call it a *Killing horizon*, after the mathematician who described such symmetries — has the structure just described: distinct events, in that order, with no separation between them. The horizon of every settled black hole is one, spinning or charged, by a theorem of the 1970s (under its standard assumptions). And collapse of any shape ends at one. A lopsided, spinning star sheds its lopsidedness as it falls, and what the outside world approaches, at the end of its time, is the horizon of the settled black hole it is becoming. With a cosmological constant in the picture, that the settled black holes are exactly that one family is expected but not yet proved.

I proved this after a couple of years of careful thinking and writing about the relationship between spacetime and what it can mean for something to physically exist, in a sense that holds together relativistically — and about what that means for gravitational collapse in particular. Through that work I realised that a theorem could be proved in standard general relativity, with nothing added to it, that overturns the idea of a black hole that physics has carried since the 1960s: an idea that reads into the mathematics something the mathematics never demanded. That theorem is the main result of the first paper in the work this explainer describes, and it reaches further than it might seem. Roger Penrose's Nobel Prize–winning singularity theorem, which guarantees a singularity inside every black hole, starts from a surface sealed inside a completed horizon. The proof is correct. But the event horizon of every collapsing object occurs, from every outside perspective, only as a singularity of the geometry placed at the end of infinite time — so the surface the proof starts from never physically arises, and the theorem has nothing in the real universe to apply to. This past spring I wrote up a final account of why this must be, and from there the rest of the picture tumbled out as I worked it through with Claude, drawing on what I'd done over those couple of years and on my doctoral thesis, completed in 2012: thinking through the mathematics, and how its pieces would all fit together into one coherent whole.

So the black hole doesn't sit there swallowing things forever, as a persisting object. What sits there is matter still collapsing toward a horizon it never completes. And what lies on the other side of that completion is not "inside the black hole" in any sense a map of our universe could show. It is something with its own time.

Where an ending turns into a beginning

Here is the central move, and it's simpler than it sounds.

Start with the simplest universe general relativity allows: one that holds nothing at all except the cosmological constant. Its geometry is called de Sitter space, and it has a shape you can picture. Draw it with time running up the page and one direction of space running across, and it is a surface like a cooling tower — narrowest at the waist, flaring out above and below. Each horizontal circle around the tower is the whole of space at one stage, smallest at the waist and growing above it. The curves that climb the tower while keeping their place on those circles are the paths of things at rest. And the surface has a remarkable property, the one a cooling tower's lattice of straight steel struts shows off: although it is curved, it is made entirely of straight lines, two families of them, each line running from the bottom of the tower to the top, tilted one way in one family and the other way in the other. In de Sitter space those straight lines are the paths of light.

What makes a path a light path is the geometry's rule for measuring separations — physicists call it the *metric*. It assigns a length to every short step across the surface, and light paths are the ones along which that length is exactly zero, the zero-length separations of the first section. The surface itself is just a set of points; the rule is laid over it. And a surface can carry more than one such rule. Keep the same points and the same circles, change the rule, and the roles of the paths change with it: one family of straight lines stops being light and becomes the paths of matter, and the climbing curves that were matter at rest become the paths of light. That exchange is the central move. Physicists call it a causal reassignment.

The first section already told us which way to make it. A collapse's slices of "now" pile up against the horizon and end lying along it, tangent to one of the light paths running in the horizon itself. The direction the collapse hands on is that light direction. The reassignment promotes it to the direction matter moves in — so what the outside universe reaches only at the end of all its time is where the new universe's own time is seeded, though, as we'll see, the universe that grows from it begins further along the same curve.

Nothing has to be added for this to work. No new field, no new force, no modification to Einstein's equations. It is the same geometry, read with a different rule for which paths are light.

And what comes out of that reading is not a universe of arbitrary shape. A family of light paths can meet a horizon in three ways. It can cut across it, which gives two separate horizons — a black hole's, and the cosmological horizon the constant draws around it — the field of an ordinary mass sitting in space. It can miss, which gives no horizon at all. Or it can graze. The collapse rules out the second, since it forms a horizon, and rules out the first, since the direction it hands on runs along its horizon rather than across it. One member is left: the one in which the black hole's horizon and the cosmological horizon meet and merge into one. Physicists have known that configuration for seventy-five years as the Nariai geometry, and long treated it as a curiosity.

Read from the inside, in the frame of the matter moving along it, that geometry is a universe that is the same everywhere and looks the same in every direction, whose size grows according to one law:

r = R sinh^(2/3)(3τ / 2α).

Here r is the radius of the universe's space, τ is cosmic time, and sinh is the hyperbolic sine — a curve that rises slowly at first and then ever faster. The law has only two constants in it, and the cosmological constant Λ sets both. One is a length, α = √(3/Λ), about 17 billion light-years: the one length the cosmological constant defines, and every special radius on the journey ahead will turn out to be a fixed multiple of it. The other, R, is about 12.4 billion light-years, and it too is set by Λ alone. That function is exactly the one cosmologists fit to our own universe's expansion — not a close approximation, the same function.

What we measure when we measure how much matter the universe holds is where along that curve we stand. That position is one number. Call it x_0: today's radius divided by the radius of the merged horizon. The fraction of the expansion that standard cosmology attributes to matter is then $\Omega = 2/(x_0^3 + 2)$, and its measured value, about 0.30, means x_0 is about 1.66. The same number fixes when the universe began to speed up: at a redshift of $x_0 - 1 = 0.66$ — that is, when the universe was 1/1.66 of its present size. Hold on to that number; it comes back.

That is the content of the claim that this universe *must* expand, and at the rate we see. The expansion isn't a leftover momentum fighting a brake. It is the shape the existing universe has, given the one constant it lives in.

And "existing" has a precise meaning here, which the rest of the story leans on. What exists is a three-dimensional space — the universe at large, finite and without edge — and it advances. How far it has advanced is cosmic time. Everything that happens as it advances, laid out all at once, is spacetime: not a second thing that exists, but the record of what happens. I'll call the existing space *the layer*.

Flat and closed at once

Now a puzzle that has bothered cosmology for a century, and that this picture turns inside out.

Measure distances across our universe, and space comes out flat: triangles add to 180 degrees on the largest scales we can test. But a universe that came from a finite collapse ought to be closed — finite, with a three-dimensional sphere for its space, the way a balloon's skin is a finite two-dimensional sphere. Which is it?

Both. And the reason why is the most useful idea in the whole picture.

Picture the universe's space as a three-sphere — the three-dimensional version of the balloon's skin, finite and without edge. It can be filled completely by a special family of circles: through every point runs exactly one of them, no two cross, and any two are linked once, like two links of a chain. Mathematicians call this the Hopf fibration. In this picture all of the universe's matter is moving along those circles, every piece at the same speed and in the same twisting sense — and the path each piece traces through the record is one of the light paths of the empty geometry, the ones the reassignment turned into the paths of matter. Since all of it moves together, no piece of matter can tell it is moving relative to any other.

Now redescribe the same situation with the matter called "at rest." In that description, space itself carries the uniform twist along those circles, and the matter sits still within it. Make that swap, and the slices on which the resting observers' clocks agree — the slices each of them would call "now" — are perfectly flat: on them, triangles add to exactly 180 degrees and distances are those of flat space, exactly. Those are the slices our measurements of distance are made on, and so flat is what our telescopes find.

The sphere hasn't gone anywhere. It is the layer, the space that exists, and it is not one of those flat slices: drawn in the observers' own chart, the layers run at 45 degrees to them. That is what it means to call this cosmology *non-synchronous*. Synchrony is what clocks agree on, and it is relative to how you move; here the slices the observers' clocks agree on are flat, and the existing layer is a different family of slices through the same record. So events on one layer are simultaneous — they happen together in the one space that exists — and yet the matter's own clocks do not agree on them. Synchrony and simultaneity come apart for the matter itself, not only for observers moving through it. The measured evenness of the expansion requires a cosmic time, but it does not by itself require the matter's clocks to keep it; this cosmology is the case where they don't.

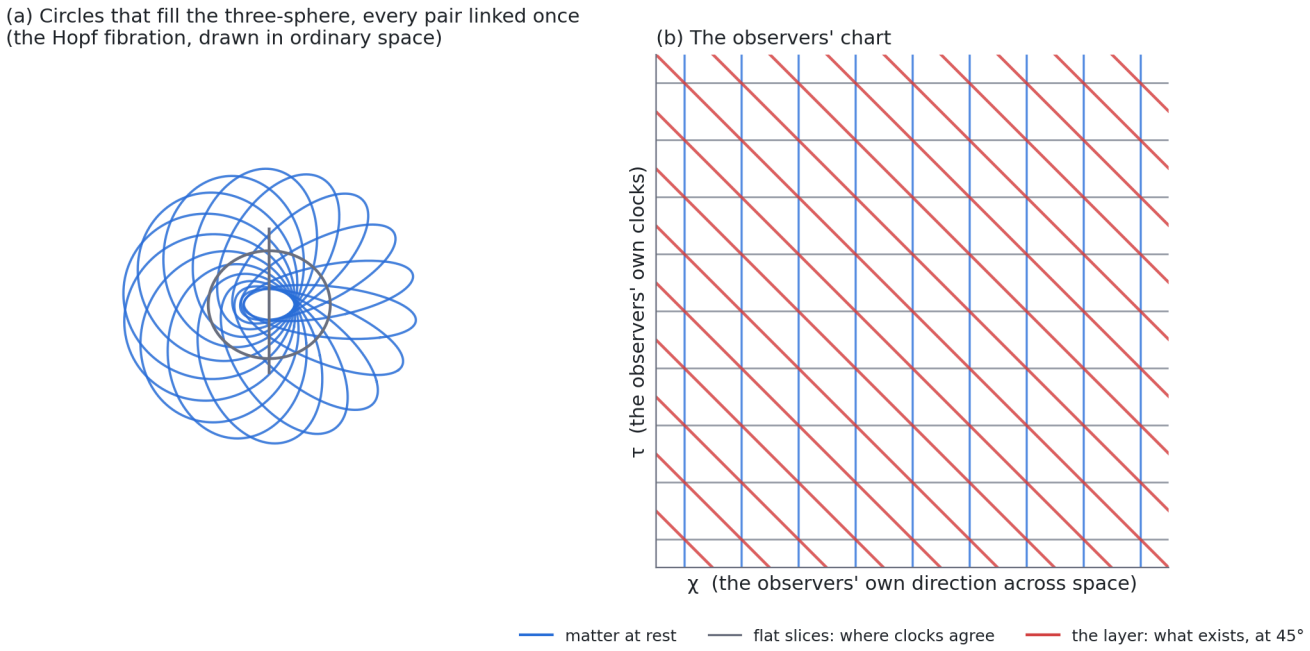


Figure 1. Flat and closed at once. (a) The three-sphere filled by circles, every pair linked once — the Hopf fibration — drawn in ordinary space; the grey circle and the straight line are two more of the same circles, the line being the one that runs through the point at infinity. (b) The same universe in the observers' own chart: blue lines are matter at rest, grey lines the flat slices on which the observers' clocks agree, and red lines the layer — the space that exists — at 45 degrees to them.

Nothing physical has changed between the two descriptions. The same matter does the same things. What changed is which family of paths we agreed to call standing still. Flat versus curved, in this universe, is nothing but that choice. And the curvature you'd expect of the closed sphere doesn't vanish when you switch to the flat description. In the empty geometry everything here is cut from,

the sphere at its narrowest has radius exactly $\alpha = \sqrt{(3/\Lambda)}$ — so the sphere's curvature and the cosmological constant are one number. Spatial curvature and the mysterious "dark energy" are one constant, read on two families of slices.

This also dissolves the oldest question anyone asks about an expanding universe: *what is it expanding into?* Nothing. The sphere isn't sitting in some larger space with room around it. It is the whole of what exists, and its growing is the growing of existence itself. The flat description and the closed description are two honest maps of that one existing thing.

Between the parent's collapse and our universe's expansion lies a stretch that neither map shows whole. Most of it runs on the cosmic clock like everything else. But one part of it does something stranger — the universe's size keeps changing while its cosmic time does not advance at all — and that is where the strangest and most interesting physics of the whole story takes place.

The lap: where a collapse stops and a universe begins

Earlier I said that, by the outside universe's reckoning, the horizon never forms. That is still true, and nothing here takes it back. But the outside's clock is not the only one the geometry keeps. The new universe's layer has a cosmic clock of its own, and on that clock nothing piles up against the end of time. The collapse runs its whole course at definite readings, through every stage a black hole is supposed to have — a horizon, an inside, a centre — each in a form the outside's reckoning never shows. That clock has a story to tell: the passage from the parent's collapse to the beginning of ours. I'll call that passage *the lap*.

Picture a single strand of that passage, and trace what its size does. Since it is always the same family of slices, we can track the whole journey by one number: the radius of the sphere that strand belongs to.

To follow it we need one rule, and it is a familiar one. A ball rolling on a hill moves so that its speed squared equals its energy minus the height of the hill where it is; where the hill rises to meet its energy, the ball stops and turns back. Every path in this geometry that falls straight in or out obeys exactly that rule, with the radius in place of position:

$\text{speed}^2 = E^2 - f(r)$, where $f(r) = 1 - 2GM/c^2r - r^2/\alpha^2$.

The function f plays the hill. It is the same function whose zeros are the horizons — the $2GM/c^2r$ term is the mass's pull, the r^2/α^2 term the cosmological constant's push — and paths come in a family labelled by their energy E. The member with no energy to spare, $E = 0$, stops exactly where $f = 0$: at the horizons. The member the cosmology is made of has $E = 1$, and it stops where $f = 1$. Keep that rule in mind; most of what follows is read off it.

The strand comes in from far away, shrinking — out of the parent universe at large, in its late and thinning age, which on this clock stretches back without limit. Out there the radius keeps the time. At about twenty billion light-years — $2\alpha/\sqrt{3}$, to be exact — it passes a zero of f. The whole lap is drawn along one curve, and the zeros of f are where that curve turns; I'll call them seams, and there are two, so I'll name each one every time. This is the *back seam*. It is the horizon the cosmological constant draws around every mass, the edge of the mass's own neighbourhood, past which the universe at large recedes out of reach. Crossing it inward, the radius changes character. Outside, it is a time: the radius is what the clock reads. Inside, it marks a place instead, as it does in the still surroundings of a star. From there it keeps falling inward for nearly fifteen billion years of cosmic time, and the matter caught in it compresses and heats as it goes.

Then, at a radius of about twelve billion light-years, the collapse stops. On the cosmic clock the shrinking simply halts. This is the *turnaround*: the place where $f = 1$, where the cosmology's member of the family runs out of speed. And it is this clock's own horizon. A horizon is where the $E = 0$ member turns; the turnaround is where the $E = 1$ member turns — the same kind of point, read by a different member of one family. On this clock there is no event horizon to cross at all, completed or not; the turnaround is what stands in its place.

What happens next — call it *the lift* — is the strangest stretch of the whole story, and the most beautiful. Inside the turnaround the rule gives a negative answer: for the cosmology's member, with $E = 1$, the speed squared $1 - f$ comes out below zero for every radius between the turnaround and zero. A ball cannot be anywhere its speed squared would be negative. That is a forbidden region — a barrier the falling matter has no ordinary way across. Quantum particles do cross such barriers; it is called tunnelling, and the mathematics of it is exact. If the speed squared is negative, then a step in time whose own square is negative balances the books — and a number whose square is negative

is an imaginary number. So across the lift, cosmic time turns imaginary. Its real part stops: not a second passes on the real clock. The radius keeps going — down past twelve billion light-years, through everything smaller, all the way to zero — while the imaginary part of time runs. How far it runs is fixed by the law itself: $\pi\alpha/3$, about eighteen billion years of imaginary time. Even the shape of the descent follows. The hyperbolic sine that governs the expansion becomes an ordinary sine when its argument is imaginary — $\sinh(ix) = i\cdot\sin x$ — so across the lift the radius follows a sine, the kind of curve it follows inside a black hole's horizon, where everywhere else on the journey it follows a hyperbola, the kind it follows outside one.

That sounds like a magic trick, so it's worth saying what it is and isn't. Imaginary time is not science fiction. It is a tool physicists have used for decades: it describes how particles tunnel through barriers they lack the energy to climb, and it is how the temperature of a black hole was first computed. Here it does the first of those jobs, and it isn't bolted on from outside: the barrier is the rule's own, and the lift crosses it the way a tunnelling particle does. Read off the one rule, the motion has three characters along the lap — real at the seams, where the strand moves at exactly the speed of light; zero at the turnaround; imaginary across the lift. Not three laws of motion: one family of paths, read at three of its natural turning points.

And the three are values of a single quantity. Each layer strings its matter together in a definite direction, strand after strand, and that direction is the same direction of time a mass's still surroundings use — the one along which a star's neighbourhood looks the same from moment to moment. Measure its squared length, the way the metric measures any step, and it comes out as $-f$: the hill itself, with its sign turned over. Where it is zero the direction is a light path, and that happens at the seams. Where it is -1 it is exactly one unit of time, the comoving clock's own, and that happens at the turnaround. Where it is -2 the speed squared is -1, the imaginary counterpart of light's speed, and that happens inside the lift. At the branch point the squared length runs out through infinity and comes back with its sign reversed, so the order within the layer turns from an order in time to an order in space. That is the radius taking up the clock, seen from the layer.

Two cautions keep this honest. First, the eighteen billion years is a length of *path*, not of *history*. Nothing ages through it, and it doesn't make anything repeat. Second, it doesn't contradict what was said about the horizon. The two reckonings are two different clocks. On the outside's, the collapse

never completes, and the horizon happens only at the end of all its time. On the cosmic clock of the new universe's own layer, the collapse reaches its own horizon — the turnaround — at a definite reading and goes on through it, and the lap is told on that clock.

At the bottom of the lift the radius reaches zero, and there the universe begins. This is the black hole's centre and the new universe's beginning at once: the place where the collapsing hole and the expanding universe meet. Drawn as a curve, the lap stands exactly vertical there — the opposite of a seam, where it turns flat — and passes straight through: a *branch point*, not a wall. What passes through with it needs saying exactly. A body falling along the collapse reaches the branch point after a finite time on its own clock and ends there, pulled apart by tides that grow without limit; what continues is the curve, not the body. What crosses is content — the pattern the collapsing universe carries, in the precise sense the next section spells out. The crossing is lossless for content and fatal for bodies. And the radius changes character once more: from here on it keeps the time, as it does inside a black hole's horizon. The lift carried the first mark of a black hole's inside, a time that stops being a time; here the radius takes up the second, and becomes the clock. That is why, in our universe, size is the clock.

Something else turns over there too. In the geometry's own terms, the radius on the collapsing side is negative and on the expanding side positive. The reflection that flips that sign, flipping the sign of the mass along with it, leaves the geometry unchanged — and it does everything charge conjugation, the operation that exchanges matter with antimatter, does, except one thing: it cannot reverse the sign of electric charge, because the geometry carries charge only as its square. That last step comes from the matter itself. So the collapse our universe came from sits on the antimatter side of the branch point. In that precise sense, our universe was born from an antimatter black hole — and its own inhabitants, looking back across the same point at us, would say exactly the same of us.

Neither side is favoured. The quantity that decides which path a system actually takes — physicists call it the action — comes out equal and opposite on the two branches, and the two sum to exactly zero. The geometry tips no scale between matter and antimatter: whatever imbalance of matter over antimatter our universe has, it brought with it.

From zero the radius grows. This is our universe's expansion, on the same cosmic clock, starting from the branch point. It grows fast at first, then ever more slowly, for about seven and a half billion years, until at about 9.8 billion light-years — $\alpha/\sqrt{3}$ — it reaches the second seam, the *front seam*, where it is growing slowest of all. Nothing about the radius's character changes here, as it did at the back seam and again at the branch point: it only touches the horizon's value and carries on. What turns instead is its pace. It stops slowing and starts speeding up, and has been accelerating ever since, for about the last six billion years.

Both seams are horizons of exactly the kind the first section described. At each, the layer itself lies along light paths, and the events each of those paths threads, one on every strand it meets, are distinct, in order, with no separation between them. The order along each path is at once the order of the strands across the layer and the order of their own clocks. So the black hole's horizon and the lap's seams are not two things that happen to resemble each other: they are one kind of structure, met in both places. The seams differ in how hard they pull — the back seam pulls and the front seam does not, as the next section spells out — and that difference matters at the front seam. With no pull, the horizon's own count of the events along its light paths is the natural, evenly spaced count a light path carries of itself. That is what lets the collapse's horizon and the cosmological horizon be one and the same horizon there, with the order handed over exactly, count and all. At the back seam the order still carries over, but the count does not.

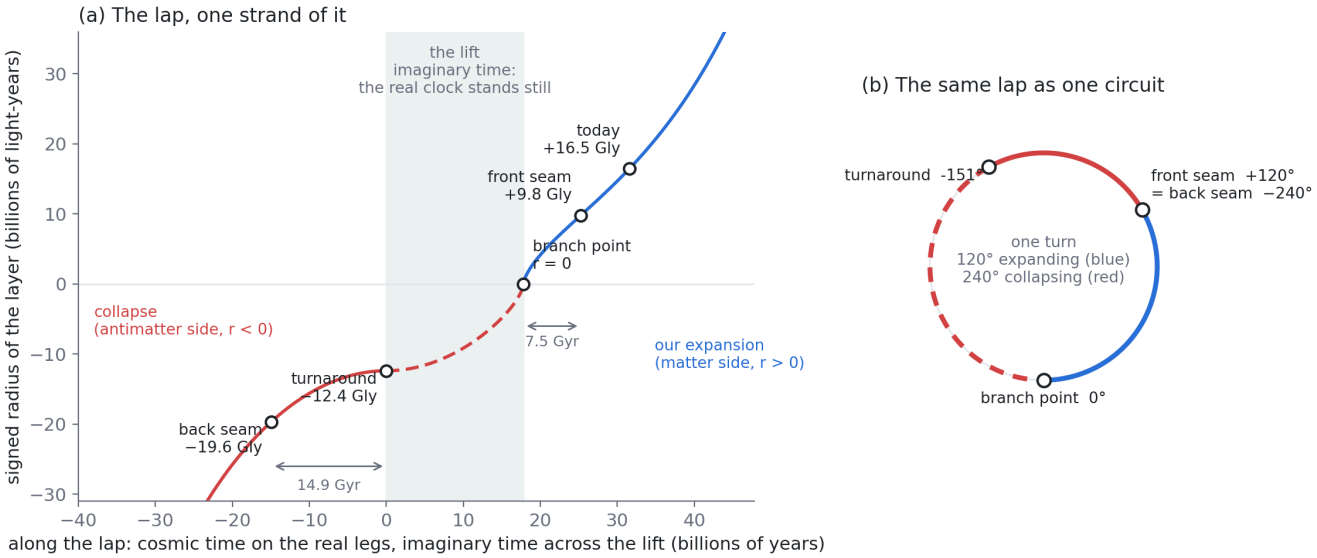


Figure 2. The lap. (a) One strand's size, signed — negative on the collapsing side, positive on ours — along the whole path. The collapse (red) comes in through the back seam and halts at the turnaround; across the lift (shaded) the time is imaginary and the real clock stands still; the radius passes through zero at the branch point, and our expansion (blue) runs out through the front seam to today. The arrows mark the two stretches in the ratio two to one. (b) The same lap as one circuit: the branch point at 0°, the turnaround at about −151°, and the front seam at +120° and back seam at −240° as one and the same point.

Two numbers frame all of this, and they are not a coincidence. From the back seam to the turnaround is about fifteen billion years of cosmic time; from the branch point to the front seam, about seven and a half: two to one — and not roughly, but exactly. Draw the whole lap as one closed circuit of the underlying geometry, and the branch point divides it into a third and two thirds, 120 degrees and 240 degrees: the same two to one. And on that circuit the two seams land on the same point. The front seam sits at +120 degrees from the branch point, the back seam at −240 degrees, a full turn apart. Their radii differ, 9.8 billion light-years on one side and 19.6 on the other, because the radius carries a sign and the two are read on opposite sides of the branch point; the place in the underlying geometry is one.

Now measure the whole journey in a different time: *conformal time*, which counts each stretch of cosmic time in units of the layer's size at that moment. Light crosses equal distances on the layer in equal amounts of it, which is why it is sometimes called the time light keeps. Measured that way, something remarkable happens: all of it fits. The collapse, all the way in from the parent's endless late age, takes only a finite stretch of conformal time; so does our expansion, all the way out into its endless future. Lay the collapse end to end with the lift, which runs at right angles to it because its

time is imaginary, and the two make the short sides of a right triangle whose long side is exactly as long as the whole of our expansion. In units of α the sides are 1.928, 3.339 and 3.855 — in the proportions 1 : √3 : 2, the 30-60-90 triangle every geometry student meets. A history without end in either direction on the cosmic clock closes, on light's clock, into one clean geometric figure. And the two to one turns up again, the other way round: on light's clock the whole of our expansion is exactly twice the whole of the collapse. Keep the lift's side in mind, 3.339; it decides what the next section is about.

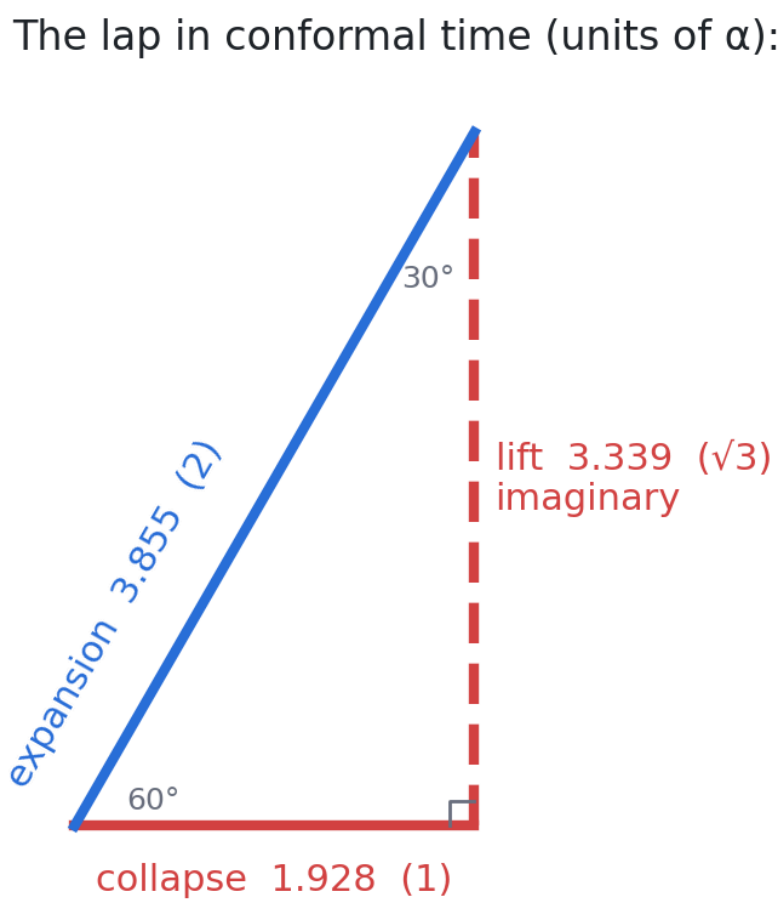


Figure 3. The lap on light's clock, conformal time, in units of α. The collapse and the lift, at right angles because the lift's time is imaginary, are the short sides of a 30-60-90 triangle whose long side is the whole of our expansion: 1.928, 3.339 and 3.855, in the proportions 1 : √3 : 2.

Most of the physics we can actually see happened on one short stretch of all this: the seven and a half billion years from the branch point out to the front seam. Over that third of the circuit the young universe was hot, glowing and ringing with sound. It forged its first nuclei and released the

light we still see as the microwave background. The rest of the lap — the parent's long collapse and the lift through imaginary time — is invisible to us directly. But it isn't silent. It decides what reaches the branch point and what doesn't, and that is where the story goes next.

What crosses the beginning

The universe's beginning is a passage, not an explosion: something goes in on one side and something comes out on the other. So the natural question is what makes it through. The answer turns out to be both very little and exactly enough — and two separate filters decide it, one acting on direction and one on time.

Size crosses. Direction does not. Think of every possible pattern the collapsing universe could carry as a sum of simpler pieces, the way physicists break a chord into its pure tones. On a sphere the pure tones come with a *degree*, L: degree 0 is the overall size; degree 1 a lopsidedness, one side bigger than the other; degree 2 a squeeze along one axis; and finer and finer ripples from there. In real time a pattern of degree L oscillates, rising and falling at a rate that grows with its degree. But across the lift time is imaginary, and in imaginary time an oscillation becomes an exponential decay: the same mathematics that describes a wave describes, with time turned imaginary, something dying away. The lift is a fixed length of conformal time — the triangle's imaginary side, 3.339 — so each pattern comes out of it multiplied by a decaying exponential of its own rate times that length. The finer it wiggles, the harder it is crushed.

The one piece that doesn't wiggle at all — the overall size — passes through untouched, exactly. The simplest lopsidedness comes out at under a tenth of its strength; the squeeze at four thousandths; and from there every further two steps in degree cost about three powers of ten — a hundred-thousandth at degree 4, a hundred-millionth at degree 6, and on down. None of these numbers was chosen; there is nothing in the geometry to adjust. One caution belongs right here. These degrees count patterns on the layer, the sphere of space itself. The patterns astronomers chart on the sky are counted differently, by angular size, and how a pattern on the layer lands on the sky is a separate step, worked out on its own — not a relabelling. And a second, independent route gives the same answer: where the parent's horizons merged, the geometry is a two-dimensional de

Sitter space with a sphere attached at every point, and de Sitter space is known — it is a theorem — to forget everything but its own scale. On it, every pattern of degree 1 or higher decays and only the overall size survives.

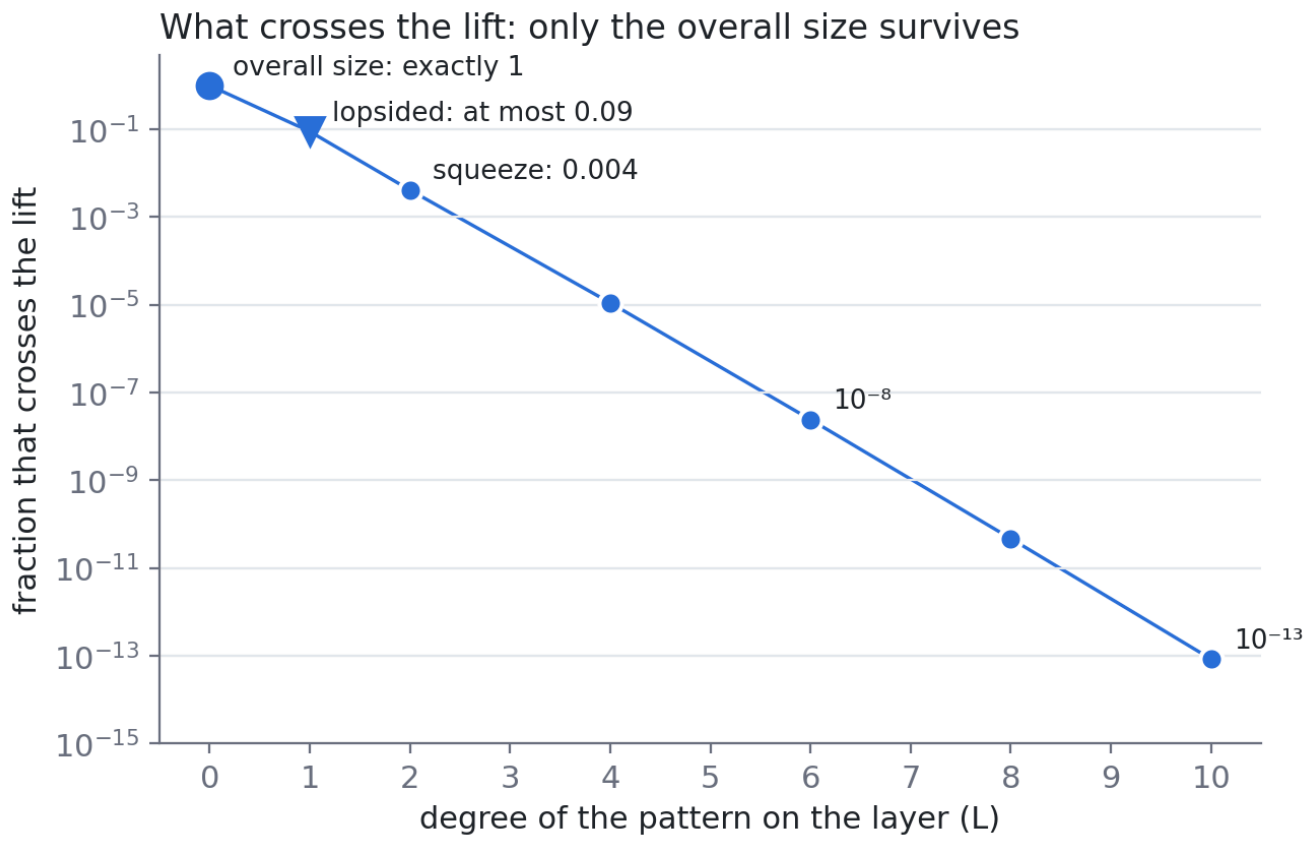


Figure 4. What crosses the lift, by degree L of the pattern on the layer. The overall size (L = 0) crosses whole; the lopsidedness (L = 1) at most 0.09; the squeeze (L = 2) 0.004; and each further two degrees costs about three powers of ten. The scale is logarithmic: each gridline is a hundredfold.

That is why the sky is smooth. One of the great puzzles of standard cosmology is that the microwave background looks the same, to one part in a hundred thousand, in every direction — including regions that, on the standard account, could never have exchanged a signal. The standard fix is inflation: a separate, adjustable burst of expansion added to stretch away the differences. Here nothing is added. The beginning itself is an isotropizer. Whatever lopsidedness the parent's collapse carried in, the passage hands on only its size. A smooth sky isn't a coincidence to be explained. It is what a universe that began this way has to look like.

The slate is melted clean. As the parent's matter falls inward on the collapse leg, it heats — eventually to at least some three thousand times the temperature at which deuterium, the most fragile nucleus, can first hold together. Whatever the parent had built, its heavy elements, its stars, its chemistry, is broken back down into free protons and neutrons. Then our side of the branch point cools back down through the same temperatures, at the ordinary rate, and ordinary nuclear physics forges the first light nuclei again, from scratch. Run that, and helium and deuterium come out at the amounts astronomers measure. The oldest stars we see are almost free of heavy elements, which is exactly what a slate wiped clean requires. Lithium comes out at the same threefold excess the standard model predicts — that puzzle sits in the nuclear physics both pictures share, not in this geometry.

The sound starts together. The second filter acts on time rather than direction. The young universe rang like a bell: pressure waves sloshing through the hot plasma, which we now see frozen into the microwave background as a series of peaks. Those peaks are sharp, and that is a clue. Sharp peaks mean every sound wave of a given wavelength started at the same moment, in step; if they had started at scattered times, the peaks would have washed out into a smear. In the standard picture that coordination is another job handed to inflation. Here it comes from the beginning itself. Anything still oscillating when it reaches the lift is damped the same way patterns are — at the scale of the first sound peak by a factor of about 10^{-66}. But nothing arrives oscillating. In the last stretch before the branch point, every pattern grows wider than light can cross in the time remaining, and a pattern light cannot cross can't ring: it freezes. Frozen patterns pass the lift untouched. So what crosses is a pattern's strength and its slight tilt — never its phase, the point a wave had reached in its rising and falling. The sound waves of the parent's collapse don't make it through, and the observed peak positions independently require that they don't. Every sound wave on our side starts fresh, together, from the branch point. One beginning, one shared starting gun.

And the parent hands over a small inheritance. Some things the geometry does not make, and the useful question is where they come from instead. The proportion of matter to light; the slight excess of matter over antimatter, since the passage itself favours neither; and the overall strength of the faint ripples that later grow into galaxies, together with their slight tilt toward larger scales — astronomers call these two the amplitude and the tilt. These come from the parent, handed across at the beginning — the way a child inherits a few particulars from a parent while the shape of a

human body comes from something deeper. Standard cosmology treats the first two the same way, as measured inputs, so nothing is lost by it. And that the tilt belongs on this list is settled by counting rather than by elimination. Trying candidates one at a time — something in the geometry that might produce a tilt after all — can never finish, because there is always another candidate to think of. Written out properly, the pattern that emerges is a product of just three things, and a tilt can only come from a term that varies with scale. One of the three is a plain constant and cannot tilt anything. That leaves exactly two, and both are already accounted for: what the collapsing interior multiplies the pattern by, which is itself a constant here, and what the parent handed over. So the tilt is the parent's not because the alternatives have run out but because there were only ever two places it could come from.

The ripples turn out to be more interesting than the standard story makes them. On the inflationary account they begin as quantum jitter in empty space, stretched to astronomical size. Here that can't be their source: the jitter this geometry's empty space supports is weaker than the observed ripples by more than a hundred orders of magnitude. The seeds of the galaxies are not quantum noise blown up. They are real, classical structure, the parent's own. And the lift, for all that it erases direction, turns out to dim them rather than wipe them out: what crosses is the parent's pattern, faint but not gone, with the broadest lopsided shapes dimmed least of all. How strong the pattern was to begin with, and its slight tilt, the geometry does not say. That is the parent's to hand over, and the work takes it as given, just as the standard account does — the one part of the ripples' story still carried as an input rather than derived.

There is one more thing on that route. Of the two seams, the back seam is a horizon in the full sense: it pulls. Every horizon has a *surface gravity* — how strongly it pulls at its own edge, the acceleration it takes to hover just outside it, as measured from far away — and Hawking showed that a horizon with surface gravity κ has a temperature, $\kappa/2\pi$ in natural units. The back seam's surface gravity is $3\sqrt{3}/4\alpha$. The front seam, the merged double horizon, only touches zero, and its surface gravity is exactly zero. And the ripples get their pattern well before they reach the back seam: in the first per cent of the fall, where the back seam sits forty-six and a half per cent of the way along it. Then they cross it. A temperature can stamp a pattern of its own on whatever lingers near it — which is exactly how the inflationary account makes its ripples. So the question was whether this one restamps the parent's.

It does not, and the reason is clean. At a horizon the wave equation has two speeds, and they are exactly 2 and exactly 0 in the geometry's natural units. Waves riding the speed-zero family skim along the horizon forever and never quite arrive; the temperature belongs to them — it is what the endless not-quite-arriving amounts to. The ripples ride the other family: they come in at speed and go through. The two never trade. Work the wave equation out at the seam and the factor the crossing applies to a wave is exactly one — not nearly one, not one for long waves, but the number one, for every wavelength. And the one scale the horizon's temperature could have stamped at all, had anything been listening, is larger than the largest wave the sky contains. Astronomers number the patterns on the sky by how finely they divide it — the number is called the multipole; the broadest patterns are numbers 2 and 3, and the first sound peak is number 220. The temperature's scale is coarser than even the simplest lopsidedness the layer can carry, and it would land on the sky at about number 4, against the first sound peak's 220: a factor of fifty-three. That factor has a twin. In position, too, the horizon stands some fifty times downstream of where the ripples get their pattern: forty-six and a half per cent of the way along the fall against the first per cent. Two different quantities, one in scale and one in place, both near fifty.

There is one more way the horizon's temperature might matter. A crossing is one thing; a horizon that simply *glows*, pouring its own radiation into the universe forming behind it, is another, and that would not care whether anything was crossing. But to work out what a horizon radiates you need its temperature to be a single definite number, and here it is not. The two seams are one place on the circuit of the lap, met a lap apart — and that one place carries two surface gravities, $3\sqrt{3}/4\alpha$ on one pass and zero on the other. A temperature has to be *the* temperature of *a* horizon; this one has two, and nothing reconciles them. The second reason is independent and just as plain: the calculation would have to compare the lap's own clock with the clock the temperature is defined in, and the conversion between those two clocks, f/r, falls to zero at exactly the two places the comparison would have to be made, because f does.

So this is not "the horizon glows and the glow turns out to be harmless." It is that the construction never hands you the object a glow would be a property of. Worth keeping the distinction: the first would be a mechanism, something to count; the second is the absence of the thing a mechanism would be about.

So the parent's pattern arrives as the parent sent it. The chain from the old universe to our sky is closed end to end, and the one thing still carried as an input is the strength and tilt the parent handed over in the first place.

Why it speeds up, and why then

In 1998 two teams of astronomers, measuring distant exploding stars, found something no one had expected. For roughly the first half of its history the universe's expansion had been slowing down, as you'd expect if gravity were pulling everything back together. Then, about six billion years ago, it stopped slowing and started to speed up, and it has been accelerating ever since. The discovery won the 2011 Nobel Prize.

The standard explanation adds a new ingredient, dark energy: a kind of energy belonging to empty space itself, whose density stays fixed while matter thins out as the universe grows. Early on matter dominates and its gravity brakes the expansion; once matter thins to twice dark energy's density, the balance tips and the expansion accelerates. That account works. But it treats the turn as the outcome of a tug-of-war between two unrelated things, and it leaves the moment of the turn as an accident of how much of each happened to be there.

In this cosmology there is no tug-of-war, because nothing is pulling and nothing is pushing.

A bend in a fixed curve. The expansion here is not a running tally of impulse against braking. It is the shape of one fixed curve — the universe's size, read on its own cosmic clock. A curve can bend one way and then the other. Think of a road that curves downhill-concave and then uphill-concave, with one point in between where it runs momentarily straight. Mathematicians call that point an inflection. The universe's expansion has exactly one. Before it, the curve bends so that growth slows; after it, so that growth quickens. At the inflection itself the universe is growing more slowly than at any other moment in its history.

And the bend is not a metaphor. Each moment's slice of the existing universe is a curved surface, and a surface's curvature can be measured from inside it: positive on a sphere, where the angles of a triangle add to more than 180 degrees; negative on a saddle, where they add to less; zero on a flat

sheet of paper. That inside-measured curvature is called the Gaussian curvature, and for the slice it is

$K = 1/\alpha^2 - GM/c^2r^3$.

The first term is the curvature of the empty geometry everything is cut from, fixed by the cosmological constant. The second is the bend the mass puts in the slice. And the universe's acceleration at each moment is exactly $r \cdot K$: that curvature, multiplied by the size. (It follows from the hill rule of the lap, differentiated once.) Written out, $r \cdot K$ is $r/\alpha^2 - GM/c^2r^2$, which has the look of the familiar Newtonian pull of a mass plus a push from the cosmological constant — which is why the standard account reads it as a tug-of-war. Here the two terms are two curvatures of one surface, not two forces. The mass bends the slice one way; the cosmological constant bends it the other; where the two cancel, the slice is momentarily flat — and that is the inflection. The slowing and the speeding up are the two sides of one bend. Nothing accelerates anything; the change from slowing to speeding up is a property of a fixed curve. And the standard account's own condition, matter at twice dark energy's density, falls at exactly the same moment. The numbers agree; what differs is what they mean.

When, and at what size. That inflection came about seven and a half billion years after the beginning, some 6.3 billion years ago — light from the galaxies we see at redshift 0.66 left them at about that moment, the $x_0 - 1$ of the second section. But here this cosmology can say something the standard one cannot even ask. In the standard picture the universe's "size" is a bookkeeping number with no absolute value; only ratios of it mean anything. Here the size is real — the radius of an actual sphere — and so the turn happened at a definite one: about 9.8 billion light-years, which is exactly one over the square root of the cosmological constant. The universe turned from slowing to speeding up when it grew past that radius.

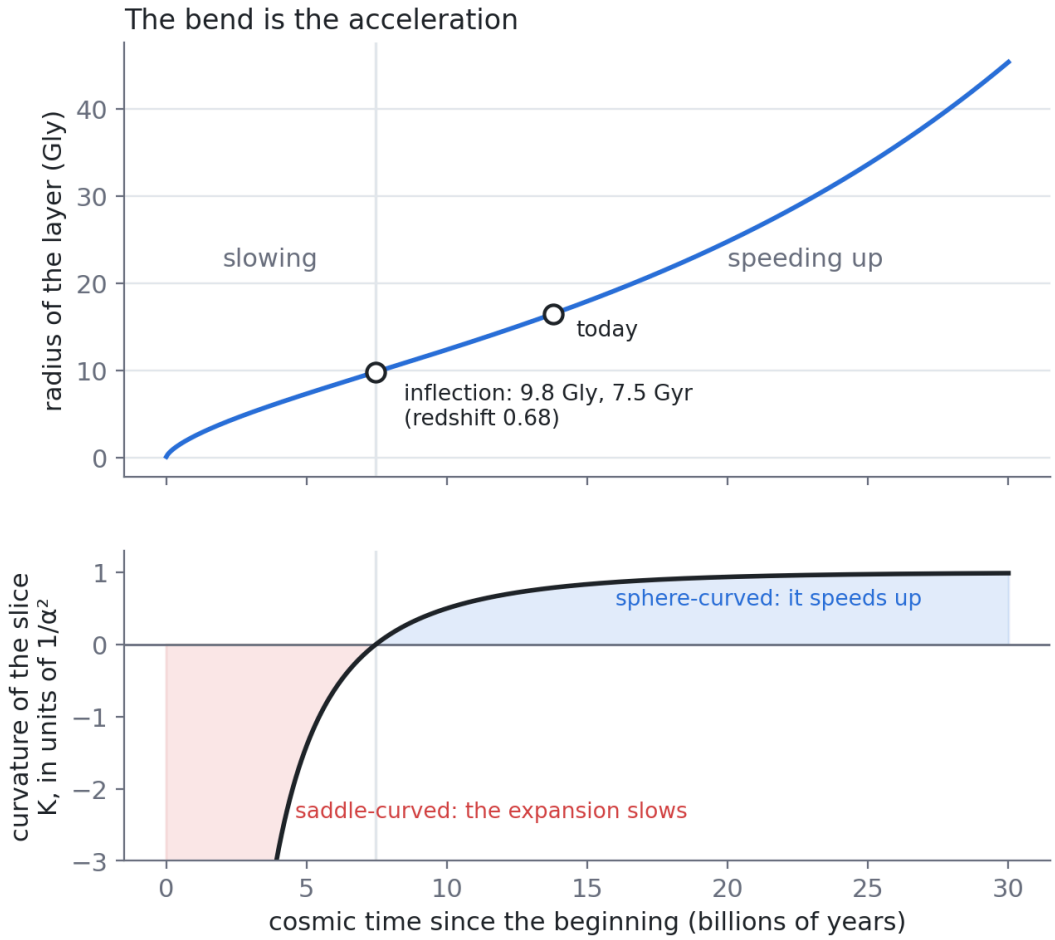


Figure 5. The bend is the acceleration. Above, the radius of the layer against cosmic time; below, the curvature K of the slice. While K is negative (saddle-curved) the expansion slows; where K passes through zero the curve inflects, at 9.8 billion light-years and 7.5 billion years; while K is positive (sphere-curved) it speeds up.

And that radius is not just any number. It turns out to be several things at once.

First: it is the universe's own holding radius. Take any mass in this universe — a galaxy, a group of galaxies, a cluster. Its gravity pulls things toward it; the cosmological constant carries things away with the general flow. There is a radius around any mass where the two balance: inside it, the mass can hold things to itself; outside it, the flow wins and carries them off. It is where the curvature K of the slice around that mass passes through zero, so it can be read straight off the formula: r = (3GM/Λc²)^(1/3). It is called the Hubble–Eddington radius, and it grows only as the cube root of the mass — eight times the mass gives twice the radius. For a galaxy like the Milky Way it is about three and a half million light-years; for our Local Group of galaxies, a little under

five million; for the great Virgo cluster, about thirty-five million. Astronomers can see this boundary around nearby groups, where galaxies stop falling inward and start receding with the flow.

Now take the one mass the geometry itself fixes — the mass at which the black hole's horizon and the cosmological horizon merge, $M = c^2/(3G\sqrt{\Lambda})$ — and put it into the same formula. Out comes exactly $1/\sqrt{\Lambda}$: 9.8 billion light-years — the radius at which the expansion turned. The universe began to accelerate at the moment it grew past the size its own mass could hold. The rule that decides whether a galaxy group holds together is the same rule that decided when the whole universe let go. The universe is the largest structure there is, and it obeys the same boundary as every smaller one — it just happens to be the one structure that lives inside its own.

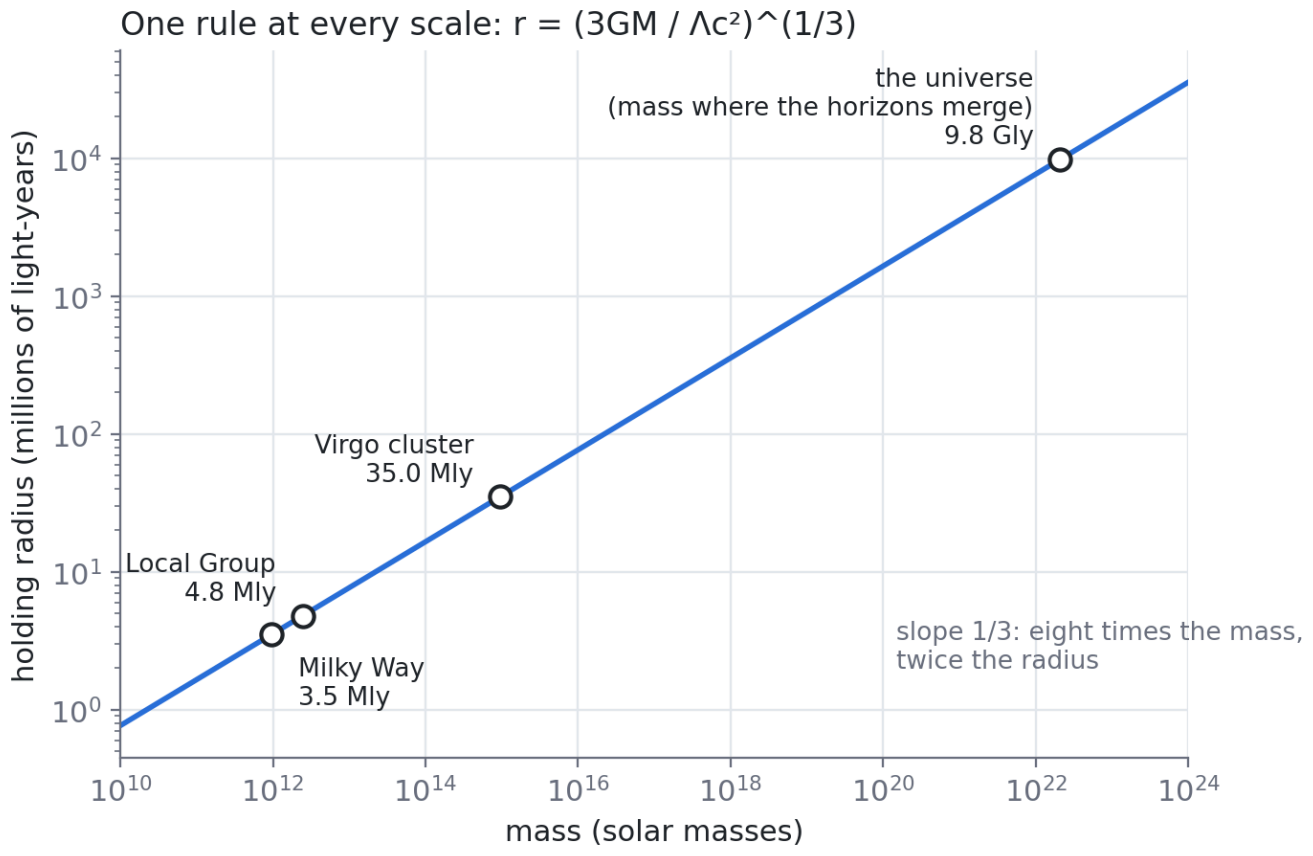


Figure 6. One rule at every scale: the holding radius r = (3GM/Λc²)^(1/3) against mass, on logarithmic axes. The Milky Way, the Local Group and the Virgo cluster sit on the same line as the universe itself, whose mass — the one at which the horizons merge — gives exactly $1/\sqrt{\Lambda}$, 9.8 billion light-years.

Second: it is the radius of the parent's merged horizon. Go back to the special configuration the parent's collapse selects — the one in which the black hole's horizon and the surrounding cosmological horizon meet and merge. They merge at a radius, and it is this one. It is also, for that same mass, the radius at which light can circle the black hole in a closed orbit, $3GM/c^2$. Three distinct properties of the parent — where its horizons merge, where light can orbit, where its gravity and the cosmological constant balance — are three formulas, and at that one mass all three give $1/\sqrt{\Lambda}$. And our universe began to accelerate as it passed through it.

Third: it is where the universe takes the horizon's shape. The universe's space is a three-sphere, carrying the uniform Hopf twist described earlier. Follow that sphere toward this radius and it doesn't stay round. It flattens along the direction of its twist, more and more: each of the Hopf circles shrinks, until in the limit every circle has closed up to a point, and what remains is a two-dimensional sphere of radius 9.8 billion light-years — the very sphere of the parent's merged horizon. What becomes of the remaining direction is just as exact: it becomes the geometry of the thin region between the two merged horizons, the configuration physicists call the Nariai throat. The throat and the universe's own space, at that radius, are one object described two ways. On its way from slowing down to speeding up, our universe passes through the precise shape of the horizon it came out of.

So the answer to "why does the universe speed up, and why then?" is this. It speeds up because the curve it lives on bends that way past a certain point, and nothing more is needed. It does so then because that point is where it outgrows the hold of its own mass — the same radius where its parent's horizons merged, and the same shape. The one constant that sets how fast the universe grows also sets the size at which a galaxy group keeps or loses its outermost members. In this picture they are not two jobs. They are one rule, read at two scales.

How far down does it reach?

So far this has been a story about the largest things there are: universes, horizons, the shape of space. It's natural to ask whether the same geometry has anything to say about the smallest — the particles everything is made of. It does, and what it says is precise. It also stops where it stops, and where it stops is as interesting as where it reaches.

What the geometry doesn't do. The great hope of a century of physics has been to find the forces of nature written into the geometry of space and time — gravity is geometry, so perhaps the strong and weak nuclear forces are too. The natural place to look in this cosmology is in the geometry's symmetries: the motions that leave it looking exactly the same, the way a sphere looks the same however it is turned. The symmetry that holds quarks together, called colour, needs six directions that can all be turned into one another. The motions of the geometry this cosmology is cut from turn six directions among themselves — but five of those are directions of space and one is time, and none of its motions can turn time into space. So colour is simply not among the motions this geometry can make. That door is shut, cleanly.

But the geometry has a second face. Turn time imaginary — the same move the lift makes, now made everywhere at once — and the difference between time and space disappears: all six directions become alike, and the same mathematics becomes a timeless, finite, perfectly round space. Colour lives there: its motions include colour's. So does the scale of the quantum, Planck's constant, which enters there through the temperature of the geometry's horizon. That face is real as structure, but nothing exists on it as a world: existing takes a cosmic time, and it has none. So the split that has dogged physics for a century — general relativity on one side, the quantum and the strong force on the other — might not be two theories waiting to be glued together. It may be one geometry, seen on its two faces: one carrying time, gravity and matter; the other carrying colour and the quantum.

What the geometry does do, for matter. Matter, though, does live on the face with time, and there the geometry delivers three things outright.

Put the simplest kind of matter particle into this geometry — the kind electrons and quarks are made of — and ask where it can sit. The radius in this geometry carries a sign, and the equation such a particle obeys depends on it. Where something an equation depends on changes sign, physicists call it a wall, and a wall of this kind has a precise effect: there is exactly one way for the particle to sit on it, and that one way spins in only one direction. It comes out with a definite handedness — a famously strange feature of real matter, which nature's laws distinguish from its mirror image. And the wall is the place where the radius changes sign, which is the very place where, read cosmologically, a universe begins.

How many walls are there? The underlying geometry has three "hinges" — three positions from which it can be sliced, set 120 degrees apart, the same 120 degrees that splits the lap into a third and two thirds. The least arbitrary way to build matter on it uses all three, so there are three walls, and three particle states, all of the same handedness. And no loop of the geometry can change that: going round any loop flips the walls' handedness only in pairs, so three of one kind can never become none. That the count also survives every smooth reshaping of the geometry that keeps the three walls is the expected behaviour of such a count, and it is traced rather than computed. Real matter comes in exactly three families — the electron and its two heavier cousins, the muon and the tau, and likewise for the quarks — and nobody knows why there are three. Here the number isn't put in. But the three walls are not themselves the three families, and the difference is worth having. The geometry carries a second three: the turnaround, read on the cosmic clock, has a threefold turn of its own, a symmetry that steps around a cycle of three. The families sit there, each a complete copy of the others, one step round the cycle. The walls fix how many there are and which way they spin; what the walls themselves index is something else, which the next paragraph meets. And there is a further surprise. The construction only works with one time and three space dimensions: a handedness needs an even number of dimensions, and the threefold count needs four or five, so four is the only number of dimensions that can carry both. In this picture, three families of matter and four-dimensional spacetime are one fact, seen from two ends.

Go one step further and let three of these particles combine. A bound state of three draws exactly one particle from each wall — so the walls index the members *inside* such a state, not copies of it — and the geometry then selects exactly the combinations nature allows for quarks — three together, or a quark with an antiquark — and forbids the ones nature forbids. It does this without supplying any force to hold them. The structure that does the selecting is flat, and in this language a force is curvature — gravity is the curvature of spacetime — so it delivers the rules of colour but not the glue.

Where it stops. It doesn't give the particles their masses — in nature those come from the Higgs field, and here they still do. On their charges it gives part and leaves part, and the line between the two is sharp. It reproduces the grouping of the left-handed particles exactly. On the right-handed side it pairs two quark states that nature keeps separate, and that split, together with the charges that follow from it, comes with the Higgs sector's content, on the ordinary route the Standard Model

already uses. What this sector adds is a rule: each family must be internally consistent on its own — free of the quantum inconsistencies physicists call anomalies — because nothing outside the walls could make up a shortfall; and the observed families meet that rule, each as a complete set. The geometry fixes the skeleton of matter — how many families, which way they spin, how they may combine, and their antimatter partners — not its flesh. The same three-way symmetry that places the horizons and the seams also places the walls where particles bind. Whether that shared root runs deeper, into the forces themselves, is a question this work has not answered.

The far horn

Now look forward. Our universe passed its front seam six billion years ago, and on its own curve it will go on accelerating forever. Its growth settles toward a steady pace: the size of the universe doubles about every twelve billion years — the length α again, now read as a time (the doubling time is the time light takes to cross α, multiplied by ln 2), and set by the one constant alone. Distant galaxies slip one by one beyond the reach of our light, and what stays bound to us is what lies inside our own holding radius. This is the far horn of the curve: the open, ever-widening region the expansion runs out into — in the language of the geometry, the de Sitter region our universe tends toward.

But our universe is not empty, and it is not quiet. Stars burn out and collapse. Galaxies grow black holes at their hearts. And now the whole story turns over.

None of those collapses ever completes a horizon — not by our clock, and not by any other way of reckoning time; on the new universe's cosmic clock, the turnaround stands where a horizon would. A completed horizon is something the actual universe never contains: what exists at every moment is matter still collapsing toward it. That is why the famous puzzles about black holes evaporating and destroying information never get started here: they presuppose a completed horizon that no moment of our world contains. That is a claim with a clear way to fail. If black holes were found to evaporate as the standard picture predicts, this account would be wrong.

But collapse, on the geometry's own terms, cannot end in nothing. It continues as a cosmology. Every black hole forming in our universe is, from its own side, the beginning of a universe: the same passage, the same lift through imaginary time, the same branch point, run again.

And there is a pleasing symmetry in it. Our universe came from the antimatter side of its branch point. By the same reflection, the universes our own black holes open onto sit on the far side of the matter–antimatter mirror from us. To their inhabitants, it is we who are made of antimatter.

So the universe is not a one-off. It is one generation in a lineage: each universe born from a collapse in its parent, growing according to the same rule, and becoming a parent in turn wherever its own matter falls together. What passes down the line is very little — the size, the smoothness and the shape come from the geometry every time; what is handed on is a small inheritance of particulars. The cosmological constant is not among them, because it is not handed down at all. A parent's collapse and its child's expansion are one curve on one geometry, and that geometry has a single scale, so every generation in the line lives in the same constant.

What the shadows are of

Go back to the question we started with. Why does the universe expand?

In the standard picture the honest answer is: because it was given an enormous start and hasn't yet been stopped. Expansion is a holdover, an impulse from the beginning still outrunning the drag of everything since. Behind that answer sits a harder question it can't touch: why there should be a universe that lasts at all, rather than one that falls back before anything has time to happen.

The picture laid out here answers differently, and the difference is the whole point. Nothing was given a push. There is one constant, present from the first instant; a layer of existence advancing through cosmic time; and the shape of one fixed curve, which is simply how that layer's size reads on its own clock. The universe doesn't expand in spite of anything. It expands because it exists, and because existing as it does, it couldn't do otherwise.

Everything else in this story has turned out to be that same fact, seen from somewhere new. The beginning: a collapse that cannot end in nothing, and so carries on as a universe. Space that reads as flat or as a finite sphere depending only on which paths we choose to call still, and a single constant that is both its curvature and its dark energy. A stretch of imaginary time that hands on a universe's size and wipes out its lopsidedness, which is why the sky is smooth. A clean slate of composition, and a single starting gun for every sound wave. A turn from slowing to speeding up at the size the universe's own mass could hold — the radius where its parent's horizons merged, and the same rule that decides whether a group of galaxies keeps its outermost members. Matter's three families and its handedness, seated on the same threefold geometry as the horizons, at the very wall where a universe begins. And a lineage: every collapse in our universe, on its own side, a new beginning.

And here is the part that still astonishes me. Wherever this picture meets the measurements, it keeps landing on the standard numbers — and every time, for a reason of its own. The expansion follows the same curve, with no push behind it and no brake against it. The light elements come out where they're observed, cooked not in a hot beginning but in a collapse's own heat, wiped clean at the turn, and cooled back through the same narrow window at the ordinary rate. The ripples come out nearly alike at every scale, not because something smoothed them flat, but because the passage they cross has no scale of its own to bend them by. And the microwave background rings with the same comb of sound peaks cosmologists have spent decades measuring — from classical structure, not quantum jitter, every wave started together at a single boundary, the spacing landing to one part in six hundred with nothing tuned to put it there. No second theory is laid over the first to make any of this happen, and no needle is pulled from a haystack of possibilities. The geometry, run forward, simply does it.

None of these pieces was added to make one of the others work. Each one turned up when the one before it was looked at closely: the same results, arriving by different roads, and the roads all turning out to be one. That is the feature I find most persuasive, more than any single result. The picture keeps paying out more than was put into it.

The test that matters, of course, is not whether a picture is pretty but whether the universe agrees with it. Here the agreement is already substantial, and to see exactly where it holds and where it does not, it helps to know what the sky's sound looks like.

The sound on the sky. Before the microwave background's light went free, the plasma that filled the universe was ringing: pressure waves of every wavelength, each rising and falling. When the universe cooled enough for atoms to form, the plasma turned transparent and the light went free, and each wavelength was caught wherever it was in its swing. The ones caught at the top or the bottom of a swing left the strongest marks. Astronomers number the patterns on the sky by how finely they divide it — the multipole, roughly 180 degrees divided by the pattern's angular size. Chart how strongly the sky varies against that number, and the strongly marked wavelengths show up as a row of peaks, evenly spaced like the teeth of a comb.

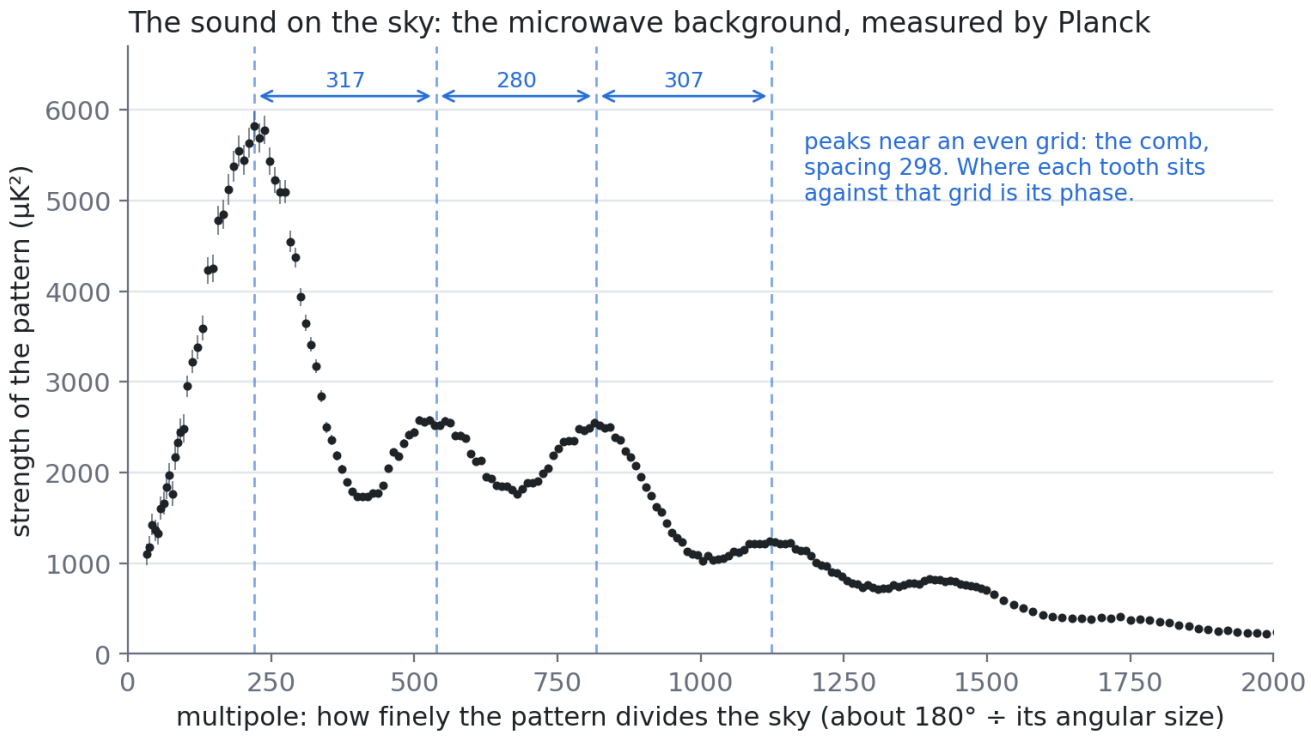


Figure 7. The sound on the sky: the microwave background's temperature pattern as measured by the Planck satellite, its strength against the multipole. The dashed lines mark the first four peaks. They sit near an evenly spaced grid — the comb, with spacing 298 — and where each one sits against that grid is the comb's phase; the gaps between neighbours differ because the phase is not the same for every tooth.

The spacing of the comb is set by one ratio. The *sound horizon* is how far sound could travel before the light went free — a ruler written into the sky. Divide the distance that light has since travelled to us by that ruler, multiply by π, and you have the spacing. And it is here, in one exact place, that this cosmology parts from the standard one. The distance is a separation across the existing layer, and it grows at the geometry's own rate, which carries no radiation. The ruler is something the

plasma itself accumulates, and it grows at the plasma's own rate, which does carry the radiation, because radiation gravitates there as it always does. The two lengths the comb is built from ride two rates that differ by exactly that one thing.

What fits. Computed that way, with nothing adjusted to fit it, the comb comes out with a spacing of 298.0 against the sky's 298.4 — right to one part in six hundred. Fixed by that spacing and the distances charted across the universe's growth, today's expansion rate comes out near 68.6 kilometres per second per megaparsec, the two routes agreeing to a twentieth of a unit, and the expansion history fits the most precise of those charts, from the Dark Energy Spectroscopic Instrument, to within the measurements' own scatter. The light elements come out where they're measured, with lithium off by the same factor of three it is in the standard account.

What doesn't yet line up is the finer shape of the peaks, and four more words make that exact. The comb's *phase* is where its teeth sit relative to an evenly spaced grid. Its *contrast* is how high the teeth stand above the gaps between them. And two effects shape both: the *driving*, the extra push the plasma's own gravity gives the oscillation while radiation dominates, and the *loading*, the extra inertia ordinary matter adds to it.

Scored as it stands, against the measurements, this cosmology's spectrum sits at 2.57 times the standard model's distance from them. Give it the same freedom the standard model is normally judged with — its expansion rate, its two densities and the tilt of the primordial ripples all fitted to the microwave data — and it sits at 1.57 times: the freedom closes about a third of the gap and leaves the rest. Nor is any parameter simply set wrong. Given its head, this spectrum settles within a twentieth of a standard deviation of the background the distance measurements had already fixed, at a cost of one part in a hundred. The sky is not pulling it anywhere. What is left over is a shape, under a per cent in size, that no free parameter absorbs.

The shape shows up in two comparisons, and each says something exact. Set against the standard model's spectrum computed by the identical procedure, the two agree on where the peaks are and differ between them: this cosmology's oscillation swings about four per cent wider about its smooth average — a difference of contrast, not of phase. It is not in the sound itself. At the moment the light went free, the two calculations' oscillations agree to half a per cent; the difference is made in the projection, in how that pattern is carried across the distance to us. Build that distance on the

plasma's rate instead of the geometry's and the difference disappears entirely, and the fit then sits slightly closer to the measurements than the standard model's own. That is the same as putting the radiation back into the rate the distance is built on — the one feature this construction removes, because radiation is no part of the geometry's own expansion. No other measurement closes that door. The light elements are forged on the plasma's rate whichever way the distance is built, and the distances measured across the nearby universe cannot see a radiation term that small. What keeps the distance on the geometry's rate is the construction itself, through one identification.

Set against the sky itself, the leftover takes the form of a slow slide of the comb's phase. Fitted as a wiggle, it repeats every 312 multipoles rather than every 298 — and that is exactly the same thing as a wiggle at the right spacing whose phase drifts, here by about 97 degrees across the range we can see, a quarter of one peak-to-peak cycle. The spacing itself is not in doubt: the drift is thirty times the spacing's own uncertainty, far too large to be a ruler slightly off.

What could make the phase slide is a question the sky can already be asked, and it is put to the peaks themselves: where each one actually falls, once one overall tilt has been divided out. Two numbers come out — how far the whole set sits shifted along, and how far the odd-numbered peaks sit out of step with the even ones — and on those two numbers the candidates part. The driving and the loading push the peaks in opposite directions and differ in strength by a factor of five, and both separate cleanly from a mere tilt. One candidate stays tied to the driving: this geometry carries two natural clocks, and a driving read on the wrong one looks the same on these two numbers, so an answer on the driving's side would mean *the driving, or the clock it is read on*.

Placed on that measurement, the sky sits within one of its own uncertainties of the spectrum this cosmology computes — on top of it — while adding or removing one full unit of the driving would put it thirteen away. So the sky is consistent with the driving the calculation already carries, and a change of one whole unit either way is ruled out.

Three limits come with that statement, and they bear on its two halves differently. The exclusion compares two calculations, so whatever they share cancels out of it; the agreement compares the sky with one calculation, and there nothing is guaranteed to cancel. First, projecting the sound onto the sky itself shifts the peaks; between two calculations that shift cancels to a part in a thousand, but against the real sky it cancels only as far as the calculation's projection is the sky's. Second, the

comb's spacing is taken as given: the agreement needs it right to four parts in a thousand, the exclusion only to five per cent. Third, one peak is hard to find: in about half of the ways this sky could have come out, the measurement finds three peaks rather than four, so the result rests on the half that yield four — and the spread inside that half matches the spread across all of them, so what the missing peak costs is trials, not accuracy.

The whole comparison rests on one identification: that the distance the projection uses is the one measured across the observer's own flat slice. That is not an inference a second argument could strengthen or weaken. Under the reassignment, the paths light takes are paths in that slice; the distance is read there because of what light is here. The one independent route that might have reached the same distance — following a bundle of light rays and watching the geometry focus it — does return the flat distance, but for the wrong rays: on them light arrives blueshifted, where the real sky is redshifted. It is a different question with a familiar answer. What remains is the shape itself.

There are sharper tests to come. One is the boundary around groups and clusters of galaxies — the holding radius — where this geometry says exactly how far a mass can keep hold of its neighbours against the flow. That one has been waiting on the sky rather than on the theory. Testing it needs two things measured on the same cluster: where the boundary falls, and how much mass lies inside it — and found independently of each other, which no existing catalogue manages. The samples that looked usable had fixed their outer edge by definition rather than measuring it, so the answer was built into the question before anyone asked it. A survey designed for the purpose changes that: about a hundred clusters, mapped out to five times the radius usually taken as a cluster's edge, with several thousand galaxy velocities in each — enough to weigh what is inside the boundary and to find the boundary from the motions themselves. It is the first sample that could return a real answer, and what it will meet is a prediction with nothing adjustable in it. Five times the cluster's edge is where that survey chose to stop; it is also, as it turns out, where this geometry puts the boundary — 5.23 times, from one line of algebra with no fitted quantity in it. The survey picked its reach for its own reasons and landed on the number the theory had been carrying all along. The claim at the root of it all needs no test of its own, because it isn't a hypothesis. That no black hole ever completes its horizon follows from general relativity itself, from the same structure every black-hole calculation already uses, read without assuming the conclusion. It changes nothing anyone outside could ever

see; what it changes is what we take to exist. What the sky can judge is what comes after: if collapse carries on as a universe, then ours should look like one from the inside — flat, smooth, expanding by one law with one constant setting its pace. That is the universe our telescopes find.

Each of the papers says plainly what it has not yet done, and what would do it. The programme also keeps one list of those open problems, its register, and the live part of that list is published beside the papers as the open frontier. The list gathers the papers' open problems in one place, and it leaves two kinds of unfinished thing to the papers on purpose. One is how precisely a finished result has been pinned down: a number that could still be computed or measured a little more tightly is a result with a margin, not a question nobody has answered. The other is a boundary a paper draws around what it set out to derive — something nobody has derived, on which nothing in the account rests, and which the paper names as such where the question arises. That is a limit the construction states about itself, not a debt it owes. Everything else a paper calls open belongs on the list. The list and the papers now agree: it carries every open problem they state and nothing they do not, and a check that runs with every change to the papers keeps them together, reading again any sentence about what is open that has been reworded before it is counted. When the two part company, either can be the stale one, and which it is has to be read rather than assumed. The list may have closed a question too early. A paper may not yet reflect what its own other pages settle. Or the list may be right to have closed a question and still have kept the weaker of the paper's two reasons for closing it — and that one matters more than it sounds. A question closed by naming every place an answer was looked for reopens each time someone thinks of a new place to look; a question closed by an argument that covers every place it could have been stays closed. Where a paper gives both, the list keeps the argument. Kept that way, the list is the index it is meant to be.

This is a picture, finally, about what the shadows are of. For a century the four-dimensional block of spacetime has been treated as the thing itself — the whole of what exists, laid out at once. But a map of everything that happens is not the same as what exists. Spacetime is the record; what exists is the layer advancing through it. The record is the layer's shadow — real, and built exactly by what the layer does, but not the thing that casts it. Read the record as a record, and the puzzles that came from mistaking it for the world begin to fall away: why there is a beginning, why space is flat, why

the sky is smooth, why the expansion turned when it did. What's left is a universe that was never in danger of failing to exist — born from collapse, growing by a rule set by one constant, and giving rise to others like itself. It expands because it is there.

A live page of *The Shadow of Existence*, revised as the work is. The eighteen papers it draws on are at shadow.cosmicave.org; source.